

Fezolinetant: A New Nonhormonal Treatment for Vasomotor Symptoms.

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Objective: To review the safety, efficacy, and tolerability of fezolinetant for the treatment of vasomotor symptoms associated with menopause. **Data Sources:** A literature search was conducted through PubMed using the following search terms: fezolinetant, ES259564, SKYLIGHT, vasomotor symptoms, and menopause. **Study Selection and Data Extraction:** Selected articles included those which described clinical studies of the pharmacology, pharmacokinetics, efficacy, safety, or tolerability of fezolinetant. **Data Synthesis:** Fezolinetant works by inhibiting neurokinin B from binding to its receptor in the hypothalamus, thereby decreasing the occurrence of vasomotor symptoms. Clinical trials have demonstrated fezolinetant is superior to placebo in decreasing the frequency and severity of vasomotor symptoms. Common adverse effects associated with the use of fezolinetant include headache and gastrointestinal disturbances, as well as elevations in liver transaminase levels. **Conclusions:** Clinical practice guidelines for the treatment of vasomotor symptoms associated with menopause recommend hormone therapy as the most effective treatment option. Risks associated with hormone use may limit the use of this option in some patients. Neurokinin B inhibitors, like fezolinetant, target the physiologic cause of vasomotor symptoms. With the approval of fezolinetant, as well as elinzanetant which is currently in phase 3 clinical trials, providers and patients have additional nonhormonal treatment options for vasomotor symptoms associated with menopause.

Pharmacokinetic evaluation of fezolinetant for the treatment of vasomotor symptoms caused by menopause.

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Vasomotor symptoms (VMS) affect the majority of menopausal women, with possible negative impact on several domains of quality of life (QoL). Although menopausal hormone therapy (MHT) represents an effective treatment, the risk-benefit profile is not favorable for every woman. Non-hormonal options are limited in number and efficacy. Fezolinetant is a novel oral non-hormonal drug recently approved for the treatment of moderate-severe VMS. It acts as an antagonist of neurokinin 3 receptor (NK3R), the main target of neurokinin B (a tachykinin over-expressed by kisspeptin/neurokinin B/dynorphin [KNDy] neurons after menopausal hypoestrogenism), involved in the modulation of the thermoregulatory hypothalamic center. Here, we report pharmacodynamics and pharmacokinetic properties of fezolinetant as well as its efficacy and safety data from available clinical trials. Fezolinetant has shown efficacy in reducing the frequency and severity of VMS with a positive impact on sleep- and health-related QoL and acceptable safety and tolerability profile. Given the limited availability of effective non-hormonal options for VMS, fezolinetant could potentially represent a game-changer for care of menopausal women, especially when relative or absolute contraindications to MHT use are present. Further studies to gain more information about the safety profile and potential extra-VMS benefits or disadvantages are warranted in real-life clinical practice.

Fezolinetant in the treatment of vasomotor symptoms associated with menopause.

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Introduction: Although international clinical practice guidelines recognize a continued role for menopausal hormone therapy (HT), particularly for symptomatic women <60 years of age or within 10 years of menopause, safety and tolerability concerns have discouraged HT use due to potential links with a perceived increased risk of hormone-dependent cancers, and an established risk of stroke and venous thromboembolism. There is therefore a need for safe, effective non-hormonal therapy for relief of menopausal vasomotor symptoms (VMS). **Areas covered:** This narrative review summarizes the dataset accrued for fezolinetant, a neurokinin-3 receptor (NK3R) antagonist in clinical development for menopause-associated VMS. **Expert opinion:** Altered signaling in neuroendocrine circuits at menopause leads to VMS wherein NK3R activity plays a key role to modulate the thermoregulatory center in a manner conducive to triggering the 'hot flash' response. Thus, a new generation of NK3R antagonists has entered clinical development to specifically target the mechanistic basis of VMS. Fezolinetant is the most advanced NK3R antagonist in terms of stage of clinical development. Results to date have demonstrated rapid and substantial reduction in VMS frequency and severity and associated improvements in health-related quality of life. NK3R antagonists offer a non-hormonal alternative to HT for the treatment of menopause-related VMS.

Fezolinetant for treatment of moderate-to-severe vasomotor symptoms associated with menopause (SKYLIGHT 1): a phase 3 randomised controlled study.

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Neurokinin 3 receptor antagonists are potential non-hormonal therapies for the treatment of vasomotor symptoms in menopausal women as options are scarce for those who cannot or do not want to take hormone therapy. Fezolinetant is one of the first non-hormonal neurokinin 3 receptor antagonists in development for the treatment of vasomotor symptoms due to menopause. This study investigated the safety and efficacy of fezolinetant for the treatment of moderate-to-severe vasomotor symptoms associated with menopause. SKYLIGHT 1 is a randomised, double-blind, placebo-controlled, 12-week, phase 3 trial with a 40-week active treatment extension. This trial was done at 97 facilities across the USA, Canada, Czech Republic, Hungary, Poland, Spain, and the UK. Women aged 40-65 years with an average of seven or more moderate-to-severe hot flashes per day were randomly assigned (1:1:1) to once-daily exact-matched placebo, fezolinetant 30 mg, or fezolinetant 45 mg. Randomisation was done using a web-based interactive response system and investigators, project team members, clinical staff, and participants were masked to treatment assignment. Coprimary endpoints were mean change in frequency and severity of vasomotor symptoms from baseline to weeks 4 and 12. The efficacy and safety analyses comprised all randomly assigned participants who received at least one dose of study drug. This trial is registered with ClinicalTrials.gov (NCT04003155) and is completed. Between July 11, 2019, and Aug 11, 2021, 2205 women were recruited of whom 175 were assigned to placebo, 176 to fezolinetant 30 mg, and 176 to fezolinetant 45 mg (175 in the placebo group, 174 in the fezolinetant 30 mg group, and 173 in the fezolinetant 45 mg received at least one dose [safety analysis set]). One participant randomly assigned to fezolinetant 45 mg received fezolinetant 30 mg in error, so the efficacy analysis set (full analysis set) consisted of 173 in the fezolinetant 30 mg group and 174 in the fezolinetant 45 mg group. 23 participants in the placebo group, 31 in the fezolinetant 30 mg group, and 13 in the fezolinetant 45 mg group discontinued treatment before week 12, mostly due to adverse events or participant withdrawal. Compared with placebo, fezolinetant 30 mg and fezolinetant 45 mg significantly reduced the frequency of vasomotor symptoms at week 4 (difference in change in least squares mean -1.87 [SE 0.42; $p < 0.001$], -2.07 [SE 0.42; $p < 0.001$]) and week 12 (-2.39 [SE 0.44; $p < 0.001$], -2.55 [SE 0.43; $p < 0.001$]). Compared with placebo, fezolinetant 30 mg and 45 mg significantly reduced the severity of vasomotor

symptoms at week 4 (-0.15 [0.06; p=0.012], -0.19 [0.06; p=0.002]) and week 12 (-0.24 [0.08; p=0.002], -0.20 [0.08; p=0.007]). Improvements in frequency and severity of vasomotor symptoms were observed after 1 week and maintained over 52 weeks. During the first 12 weeks, treatment-emergent adverse events occurred in 65 (37%) of 174 women in the fezolinetant 30 mg group, 75 (43%) of 173 in the fezolinetant 45 mg group, and 78 (45%) of 175 in the placebo group. The incidence of liver enzyme elevations was low (placebo n=1; fezolinetant 30 mg n=2; fezolinetant 45 mg n=0) and these events were generally asymptomatic, transient, and resolved while on treatment or after treatment discontinuation. Data support the clinical use of fezolinetant as a non-hormonal treatment for vasomotor symptoms associated with menopause. The study was placebo-controlled for 12 weeks followed by a 40-week blinded extension to assess the maintenance of effect. Furthermore, the population studied was diverse and representative of the potential target population for fezolinetant therapy. Further characterisation of the benefit of fezolinetant on quality of life, including on symptoms of mood and sexual wellbeing, merits investigation. Astellas Pharma.

Fezolinetant: First Approval.

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Fezolinetant (VEOZAH™) is an oral, small molecule, neurokinin 3 receptor (NK3R) antagonist, which is being developed by Astellas Pharma Inc. for the treatment of moderate to severe vasomotor symptoms (VMS) or hot flashes due to menopause. Inhibiting NK3R-mediated signalling in the central nervous system is a non-hormonal strategy to modulate the activity of neurones that are associated with thermoregulation, thereby reducing the frequency and severity of VMS. Fezolinetant received its first approval in the USA in May 2023 for the treatment of moderate to severe VMS due to menopause. This article summarizes the milestones in the development of fezolinetant leading to this first approval in this indication.

Treatment of Menopausal Vasomotor Symptoms With Fezolinetant, a Neurokinin 3 Receptor Antagonist: A Phase 2a Trial.

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The thermoregulatory center in the hypothalamus is stimulated by neurokinin 3 receptor (NK3R) activation and inhibited by estrogen-negative feedback. This balance is disrupted in menopause, producing vasomotor symptoms (VMSs). To evaluate safety and efficacy of the NK3R antagonist fezolinetant in menopausal VMSs. Twelve-week, double-blind, randomized, placebo-controlled study. Eight Belgian centers from September 2015 to October 2016. Generally healthy menopausal women aged 40 to 65 years with moderate/severe VMSs. Subjects were randomized (1:1) to 90 mg of fezolinetant twice daily or placebo for 12 weeks. Subjects captured VMS severity and frequency using an electronic diary. The primary outcome was change from baseline to week 12 in total VMS score with fezolinetant vs placebo. Secondary outcomes included timing of changes in frequency and severity of moderate/severe VMSs and quality-of-life assessments at weeks 4, 8, and 12. Pharmacodynamic and pharmacokinetic effects were assessed, as were safety and tolerability. Of 122 subjects screened, 87 were randomized and 80 (92%) completed the study. At week 12, fezolinetant significantly reduced total VMS score vs placebo (-26.5 vs -12.2, $P < 0.001$) and decreased mean frequency of moderate/severe

VMSs by five episodes per day vs placebo. Severity and frequency of moderate/severe VMSs were reduced from the first day of treatment. Improvements were achieved in all quality-of-life measures. Fezolinetant was well tolerated. The most common fezolinetant-related adverse event was gastrointestinal disorder ($n = 6$). Fezolinetant rapidly and significantly reduced moderate/severe VMSs, supporting its potential as an effective nonhormonal treatment option for menopausal women.

Efficacy and safety of fezolinetant for vasomotor symptoms in postmenopausal women: A systematic review and meta-analysis of randomized controlled trials.

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Vasomotor symptoms (VMS), such as hot flashes and night sweats, are highly prevalent and burdensome for women experiencing menopausal transition. Fezolinetant, a selective neurokinin 3 receptor (NK3R) antagonist, is a potential therapeutic option for mitigating VMS. Our aim is to assess the efficacy and evaluate the safety profile of fezolinetant compared with placebo in post-menopausal women suffering from VMS, by pooling all the relevant data and reflecting the most current evidence. An extensive literature search was performed in the PubMed, Medline and Cochrane Library databases from inception until June 2023 to identify relevant trials. Mean differences (MDs) and 95% confidence intervals (CIs) were calculated for continuous outcomes. Risk ratios (RRs) were calculated for dichotomous outcomes. All statistical analyses were performed using R Statistical Software. A total of six randomized controlled trials were added. For the frequency of daily VMS, the combined pooled result favored the fezolinetant group over placebo (MD -2.38, 95% CI -2.64 to -2.12; $P < 0.001$, $I^2 = 0\%$). For the severity of daily VMS, fezolinetant was again found to be superior to the placebo group (MD -0.40, 95% CI -0.51 to -0.29; $P < 0.001$, $I^2 = 70\%$). Fezolinetant (120 mg) consistently demonstrated a significant reduction in the severity of daily moderate/severe VMS compared with other doses at both 4 and 12 weeks. Patient-reported outcomes (PROs) of Greene Climacteric Scale (GCS), PROMIS the Sleep Disturbance Short Form 8b and Menopause-Specific Quality of Life (MENQoL) scores indicated significant improvement with fezolinetant. No significant difference in efficacy of fezolinetant at 4 and 12 weeks were observed in any outcome. As for safety, no significant differences in the treatment emergent adverse events at 12 weeks were found between fezolinetant and placebo. Our study significantly favors fezolinetant over placebo regarding the primary efficacy outcomes of daily moderate to severe VMS frequency and severity, including PROs, while both the groups are comparable in terms of treatment emergent adverse events. Further studies are needed to confirm these findings.

In Vitro Evaluation of CYP-Mediated Metabolism of Fezolinetant and Pharmacokinetic Interaction Between Fezolinetant and Fluvoxamine in Healthy Postmenopausal Smokers and Nonsmokers.

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Fezolinetant is an oral, nonhormonal, neurokinin 3 receptor antagonist treatment option for moderate to severe vasomotor symptoms associated with menopause. An in vitro study using human recombinant cytochrome P450 (CYP) enzymes and human liver microsomes showed that fezolinetant is metabolized to its major but inactive metabolite, ES259564, predominantly through CYP1A2, with minor contributions from CYP2C9 and CYP2C19. The clinical impact of CYP1A2 inhibition and induction on single-dose

pharmacokinetics of fezolinetant was assessed in an open-label, single-sequence, phase 1 study in healthy postmenopausal women, where the impact of fluvoxamine, a strong CYP1A2 inhibitor, and smoking, a moderate CYP1A2 inducer, were evaluated. In total, 18 participants, 9 of whom were smokers, were enrolled. Fezolinetant pharmacokinetics were evaluated after a single 30-mg dose on Day 1 and Day 7. Fluvoxamine 50 mg was administered as a single dose on Days 3 and 10 and twice daily from Days 4 to 9. Fluvoxamine increased geometric mean ratio of fezolinetant maximum plasma concentrations (C_{max}) and area under the curve from time of dosing extrapolated to infinity (AUC_{inf}) to 182% and 939%, respectively, while ES259564 C_{max} decreased to 20.1% with no significant change in AUC. In smokers versus nonsmokers, when fezolinetant was administered alone, fezolinetant C_{max} and AUC_{inf} decreased to 71.7% and 48.3%, respectively, while ES259564 C_{max} increased to 130.2% and AUC_{inf} decreased to 81.8%. A single oral 30-mg dose of fezolinetant was considered safe and well tolerated when co-administered with fluvoxamine in healthy postmenopausal women.

Effectiveness and safety of fezolinetant in alleviating vasomotor symptoms linked to Menopause.: A systematic review and Meta-Analysis.

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Vasomotor symptoms (VMS) are the most common symptoms during menopause including hot flashes and night sweats. They are highly disruptive to the quality of life. Fezolinetant is an FDA-approved non-hormonal selective neurokinin3 receptor antagonist for the treatment of VMS. In this study, we aim to assess the efficacy and safety of fezolinetant for VMS associated with menopause. Databases were searched until September 2023 for relevant studies comparing fezolinetant against placebo. Data was extracted into an online form and analyzed using RevMan (Version 5.4.1). The GRADE approach was conducted to evaluate the quality of evidence regarding efficacy outcomes. We included randomized controlled trials (RCTs) comparing fezolinetant to placebo in postmenopausal women experiencing VMS. Exclusion criteria comprised studies involving participants with contraindications to fezolinetant or those evaluating its efficacy for indications other than VMS associated with menopause. Six studies were included in this study involving 3301 patients. Compared to placebo, fezolinetant reduced the frequency of VMS episodes from baseline (SMD = -0.64, 95 % CI [-0.77, -0.5]) and (SMD = -0.63, 95 % CI [-0.72, -0.53]) at weeks 4 and 12 respectively. Additionally, fezolinetant reduced VMS severity score (SMD = -0.59, 95 %CI [-0.77, -0.42]) and (SMD = -0.4, 95 % CI [-0.54, -0.27]) at weeks 4 at 12 respectively. These reductions were positively reflected on Menopause specific quality of life score (SMD = -0.46, 95 %CI [-.57, -0.34]), (SMD = -0.37, 95 %CI [-0.48, -0.25]) at weeks 4 and 12 respectively. Regarding safety analysis, fezolinetant showed increased risk for drug-related TEAEs (RR = 1.47, 95 %CI [1.06,2.04]), serious TEAEs (RR = 1.67, 95 %CI [1.09,2.55]), fatigue (RR = 4.05, 95 %CI [1.27,12.88]), arthralgia (RR = 2.83, 95 %CI [1.02,7.8]) and ALT or AST > 3 times (RR = 2, 95 %CI [1.12,3.57]), with no other statistically significant difference regarding other safety terms. Fezolinetant has demonstrated efficacy in reducing the frequency and severity of VMS in postmenopausal women, leading to an improvement in their quality of life. These findings suggest that Fezolinetant may serve as a viable alternative to hormonal therapy for managing VMS.

Neurokinin 3 receptor antagonists for menopausal vasomotor symptoms, an appraisal.

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Fezolinetant is a neurokinin 3 receptor antagonist under investigation for treatment of menopausal symptoms. In a recent study, Lederman and colleagues¹ reported the safety and efficacy of fezolinetant for the treatment of moderate-to-severe vasomotor symptoms associated with menopause.
